Introducción a la variable compleja

¡Bienvenid@s! Números complejos (repaso). Funciones de variable compleja. Límite y continuidad. Diferenciabilidad y funciones analíticas. Integración en el campo complejo. Sucesiones y series.

Manuel Carlevaro

Departamento de Ingeniería Mecánica

Centro de Investigación en Mecánica Experimental y Computacional

manuel.carlevaro@gmail.com

¡Bienvenid@s!

Docentes:

- Profesor: Manuel Carlevaro
- Ayudante de Primera: Christian Molina

Clases:

- Miércoles de 17:00 a 19:15
- Teoría + práctica
- Clases de consulta:
  - MC: Miércoles 15 a 16 hs.
  - CM: A coordinar.

Evaluación:

- Parciales 2 (+ 2 recuperatorios) + Instancia evaluativa complementaria (febrero 2027)
- Aprobación directa: 2 parciales (≥ 6)
- Aprobación no directa - examen final: al menos un parcial con 4 o 5

Programa

1. Introducción a la variable compleja (2 clases)
2. Transformada y serie de Fourier (2 clases)
3. Transformada de Laplace (1 clase)
4. Errores en el cálculo numérico (1 clase)
5. Cálculo de raíces: soluciones de una variable (2 clases)
6. Normas de vectores y matrices (2 clases)
7. Autovalores y autovectores (2 clases)
8. Resolución de sistemas de ecuaciones lineales (2 clases)
9. Aproximación discreta y continua por el método de los mínimos cuadrados (2 clases)
10. Resolución de problemas de valor inicial (2 clases)
11. Resolución de problemas de contorno (2 clases)
12. Resolución de ecuaciones diferenciales de segundo orden mediante diferencias finitas (2 clases)

Programa

1. Introducción a la variable compleja (2 clases)
2. Transformada y serie de Fourier (2 clases)
3. Transformada de Laplace (1 clase)
4. Errores en el cálculo numérico (1 clase)
5. Cálculo de raíces: soluciones de una variable (2 clases)
6. Normas de vectores y matrices (2 clases)
7. Autovalores y autovectores (2 clases)
8. Resolución de sistemas de ecuaciones lineales (2 clases)
9. Aproximación discreta y continua por el método de los mínimos cuadrados (2 clases)
10. Resolución de problemas de valor inicial (2 clases)
11. Resolución de problemas de contorno (2 clases)
12. Resolución de ecuaciones diferenciales de segundo orden mediante diferencias finitas (2 clases)

Primer parcial: 10/6,
recuperatorios: 24/6 y 8/7.

Segundo parcial: 4/11,
recuperatorios: 18/11 y 2/12.

Tercera instancia evaluativa: 24/2/2027.

Recursos

Específicos de Cálculo Avanzado:

- CVG: Cálculo Avanzado
- GitHub: https://github.com/manuxch/calculo_avanzado

Calendario de Goolge

Recursos adicionales:

- Python y programación: https://github.com/gmg-utn/compTools

Bibliografía

- E. Kreyszig, H. Kreyszig y E.J. Norminton. ***Advanced Engineering Mathematics.*** Hoboken, USA: John Wiley & Sons, Inc, 2011.
- M.R. Spiegel et al. ***Variable compleja.*** Mexico: McGraw-Hill, 1991.
- J.W. Brown y R.V. Churchill. ***Variable compleja y aplicaciones.*** Madrid, España: McGraw-Hill Interamericana, 2004.
- P.V. O'Neil. ***Matemáticas avanzadas para ingeniería (7a. ed.)*** Mexico: Cengage Learning, 2015.
- C. H. Edwards y D.E. Penney. ***Ecuaciones diferenciales y problemas con valores en la frontera.*** México: Pearson Educación, 2009.
- R.L. Burden, D.J. Faires y A.M. Burden. ***Análisis numérico.*** 10.ª ed. Mexico: Cengage Learning, 2017.
- R.K. Gupta. ***Numerical Methods. Fundamentals and Applications.*** Cambridge University Press, 2019.
- Gilbert Strang. ***Linear Algebra and Its Applications, 4th Edition.*** 4th. Brooks Cole, 2006.
- D. Pine. ***Introduction to Python for Science and Engineering.*** Florida, USA: CRC Press, 2019.
- Facundo Batista y Manuel Carlevaro. ***Python para Ciencia y Tecnología.*** Autoedición, 2025. https://libropython.science/

¿Preguntas, dudas, comentarios?

Los números complejos

Sistema de enteros:

$$2x = 3$$
$$x = ?$$

Números "reales": $\{x : x^2 \geq 0\}$

$$x^2 = -1$$
$$x = ?$$

Motivación: $x^2 + 1 = 0$ ¿tiene solución?
Ejemplo: usar $y = e^{rx}$ para resolver:

$$y'' + y = 0$$

Los números complejos

Sistema de enteros:

$$2x = 3$$
$$x = ?$$

Números "reales": $\{x : x^2 \geq 0\}$

$$x^2 = -1$$
$$x = ?$$

Motivación: $x^2 + 1 = 0$ ¿tiene solución?
Ejemplo: usar $y = e^{rx}$ para resolver:

$$y'' + y = 0$$

$$r^2 e^{rx} + e^{rx} = 0$$

$$\therefore r^2 + 1 = 0 \therefore r = \pm\sqrt{-1} = \pm i$$

$$\therefore y = e^{ix} \text{ o } y = e^{-ix}$$

$$\boxed{y = \cos x \text{ o } y = \operatorname{sen} x}$$

De "alguna manera" i debe existir y e^{ix} debe estar relacionado a $\operatorname{sen} x$ y $\cos x$.

Los números complejos

Sistema de enteros:

$$2x = 3$$
$$x = ?$$

Números "reales": $\{x : x^2 \geq 0\}$

$$x^2 = -1$$
$$x = ?$$

Motivación: $x^2 + 1 = 0$ ¿tiene solución?
Ejemplo: usar $y = e^{rx}$ para resolver:

$$y'' + y = 0$$

$$r^2 e^{rx} + e^{rx} = 0$$
$$\therefore r^2 + 1 = 0 \therefore r = \pm\sqrt{-1} = \pm i$$
$$\therefore y = e^{ix} \text{ o } y = e^{-ix}$$
$$\boxed{y = \cos x \text{ o } y = \operatorname{sen} x}$$

De "alguna manera" i debe existir y e^{ix} debe estar relacionado a $\operatorname{sen} x$ y $\cos x$.

El sistema de números complejos:

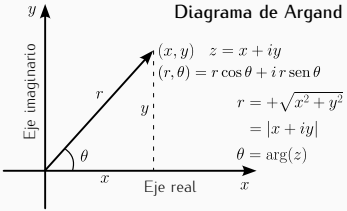
$$\mathbb{C} = \{x + iy : x \text{ y } y \text{ son reales.}\}$$

con la estructura:

$$(1)\ x_1 + iy_1 = x_2 + iy_2 \Leftrightarrow$$
$$x_1 = x_2 \text{ y } y_1 = y_2$$

$$(2)\ (x_1 + iy_1) + (x_2 + iy_2) =$$
$$= (x_1 + x_2) + i(y_1 + y_2)$$
$$(3)\ r(x + iy) = rx + iry$$
$$r \text{ real.}$$

$\therefore$ los números complejos son un espacio vectorial por definición.



Estructura adicional de $\mathbb{C}$:

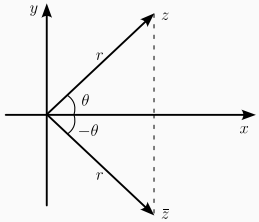
$$(4)\ (a+ib)(c+id) = \\ = (ac-bd)+i(bc+ad)$$

Caso especial:

$$(a+ib)(a-ib) = a^2+b^2 \geq 0 \\ = |a+ib|^2$$

Definición: el complejo conjugado de $z = x+yi$ es

$$\bar{z} = x - yi$$



$$\frac{c+di}{a+bi} = \left(\frac{c+di}{a+bi}\right)\left(\frac{a-bi}{a-bi}\right) \\ = \frac{(ac+bd)+(ad-bc)i}{a^2+b^2}$$

$$\frac{3+2i}{4+i} = \frac{(3+2i)(4-i)}{(4+i)(4-i)} \\ = \frac{14+5i}{17} \\ = \frac{14}{17}+\frac{5}{17}i$$

$$\therefore \frac{\text{complejo}}{\text{complejo}} = \text{complejo}$$

(excepto para división por cero).

Producto en coordenadas polares:

$$(r_1,\theta_1)(r_2,\theta_2) = (r_1\cos\theta_1 + ir_1\operatorname{sen}\theta_1) \\ (r_2\cos\theta_2 + ir_2\operatorname{sen}\theta_2) = \\ r_1r_2(\cos\theta_1\cos\theta_2 - \operatorname{sen}\theta_1\operatorname{sen}\theta_2)+ \\ ir_1r_2(\operatorname{sen}\theta_1\cos\theta_2 + \operatorname{sen}\theta_2\cos\theta_1) = \\ r_1r_2\cos(\theta_1+\theta_2) + ir_1r_2\operatorname{sen}(\theta_1+\theta_2) = \\ \boxed{(r_1r_2, \theta_1+\theta_2)}$$

Por inducción:

$$(r_1,\theta_1)\cdots(r_n,\theta_n) = \\ (r_1\cdots r_n, \theta_1+\cdots+\theta_n)$$

Caso especial:

$$(r,\theta)^n = (r^n, n\theta)$$

$$\therefore r = 1 \to (1,\theta)^n = (1, n\theta)$$

Teorema de De Moivre:

$$(\cos\theta + i\,\mathrm{sen}\,\theta)^n = \cos n\theta + i\,\mathrm{sen}\,n\theta$$

Ejemplo:

$$\begin{aligned}(\cos\theta + i\,\mathrm{sen}\,\theta)^2 &= \cos 2\theta + i\,\mathrm{sen}\,2\theta \\ &= (\cos^2\theta - \mathrm{sen}^2\,\theta) + i2\,\mathrm{sen}\,\theta\cos\theta\end{aligned}$$

$$\begin{aligned}\therefore\ \mathrm{sen}\,2\theta &= 2\,\mathrm{sen}\,\theta\cos\theta \\ \cos 2\theta &= \cos^2\theta - \mathrm{sen}^2\,\theta\end{aligned}$$

Raíces: encontrar $\sqrt[6]{i}$

$$\sqrt[6]{i} = x + iy \to i = (x+iy)^6 = 0 + 1i$$

$$\begin{aligned}\therefore\ & x^6 + 15x^4(iy)^2 + 15x^2(iy)^4 + (iy)^6 \\ & + 6x^5(iy) + 20x^3(iy)^3 + 6x(iy)^5\end{aligned}$$

Sistema complicado a resolver:

$$\begin{cases} x^6 - 15x^4y^2 + 15x^2y^4 - y^6 = 0 \\ 6x^5y - 20x^3y^3 + 6xy^5 = 1 \end{cases}$$

Teorema de De Moivre:

$$(\cos\theta + i\,\mathrm{sen}\,\theta)^n = \cos n\theta + i\,\mathrm{sen}\,n\theta$$

Ejemplo:

$$\begin{aligned}(\cos\theta + i\,\mathrm{sen}\,\theta)^2 &= \cos 2\theta + i\,\mathrm{sen}\,2\theta \\ &= (\cos^2\theta - \mathrm{sen}^2\,\theta) + i2\,\mathrm{sen}\,\theta\cos\theta\end{aligned}$$

$$\begin{aligned}\therefore\ \mathrm{sen}\,2\theta &= 2\,\mathrm{sen}\,\theta\cos\theta \\ \cos 2\theta &= \cos^2\theta - \mathrm{sen}^2\,\theta\end{aligned}$$

Raíces: encontrar $\sqrt[6]{i}$

$$\sqrt[6]{i} = x + iy \to i = (x+iy)^6 = 0 + 1i$$

$$\begin{aligned}\therefore x^6 + 15x^4(iy)^2 + 15x^2(iy)^4 + (iy)^6 &= 0 \\ + 6x^5(iy) + 20x^3(iy)^3 + 6x(iy)^5 &= 1\end{aligned}$$

Sistema complicado a resolver:

$$\begin{cases} x^6 - 15x^4y^2 + 15x^2y^4 - y^6 = 0 \\ 6x^5y - 20x^3y^3 + 6xy^5 = 1 \end{cases}$$

En coordenadas polares:

$$i = (1, \pi/2) \therefore \sqrt[6]{i} = (r, \theta)$$

$$\rightarrow i = (r, \theta)^6 = (r^6, 6\theta)$$

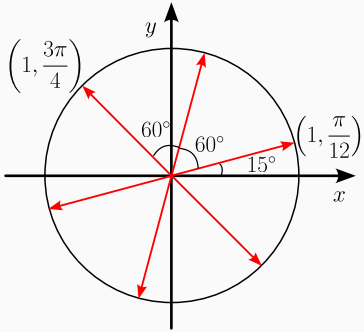
$$\therefore r = 1, \quad 6\theta = \frac{\pi}{2} + 2\pi k = \frac{1+4k}{2}\pi$$

$$r = 1,$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \frac{21\pi}{12}, \frac{25\pi}{12}, \ldots$$

$$\left(1, \frac{3\pi}{4}\right) = \cos\frac{3\pi}{4} + i \operatorname{sen}\frac{3\pi}{4}$$

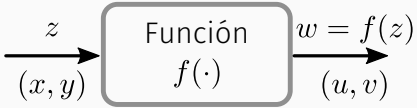
$$= \frac{1}{\sqrt{2}}(-1 + i)$$

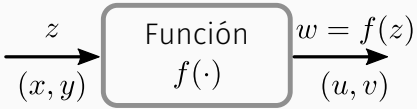


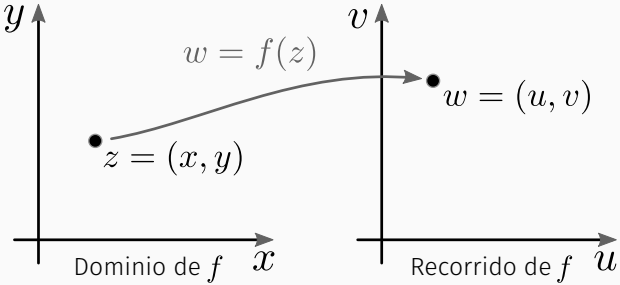
Sistema de números complejos:

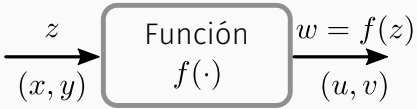
Los números complejos son **cerrados** respecto de la radicación.

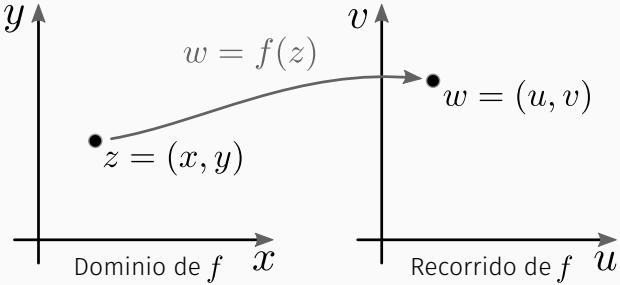
Pausa para resolver problemas: 1 – 8.


z
(x, y)
Función
$f(\cdot)$
$w = f(z)$
(u, v)


z
(x, y)
Función
$f(\cdot)$
$w = f(z)$
(u, v)

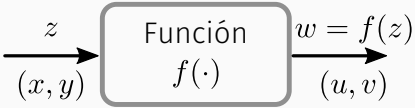

y
x
Dominio de f
$w = f(z)$
$z = (x, y)$
v
u
Recorrido de f
$w = (u, v)$

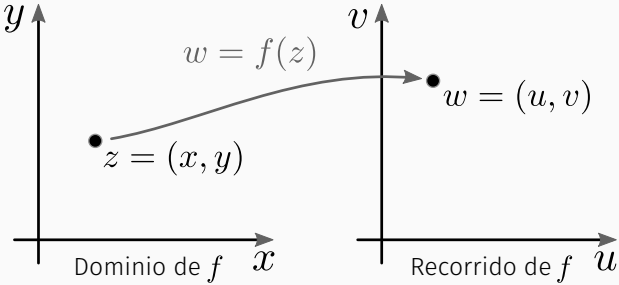




Ejemplo:

$$
\begin{aligned}
f(z) &= z^2 = (x + iy)^2 \\
&= x^2 + 2xiy + i^2y^2 = x^2 - y^2 + 2ixy \\
\therefore f(x, y) &= (x^2 - y^2, 2xy)
\end{aligned}
$$





Ejemplo:

$$
\begin{aligned}
f(z) = z^2 &= (x + iy)^2 \\
&= x^2 + 2xiy + i^2y^2 = x^2 - y^2 + 2ixy \\
\therefore f(x, y) &= (x^2 - y^2, 2xy)
\end{aligned}
$$

$\therefore f(z) = z^2$ es equivalente al sistema real:

$$
\begin{cases}
u = x^2 + y^2 \\
v = 2xy
\end{cases}
$$

Límites

$\mathbb{C}$: números complejos

$f : \mathbb{C} \mapsto \mathbb{C}, a \in \mathbb{C}$

Definición:

$$\lim_{z \to a} f(z) = L$$

dado $\epsilon > 0$ existe $\delta > 0$ tal que

$$0 < |z - a| < \delta \rightarrowtail |f(z) - L| < \epsilon$$

Límites

$\mathbb{C}$: números complejos

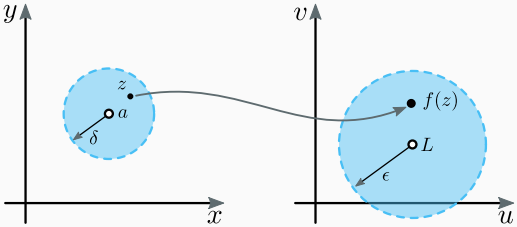
$f : \mathbb{C} \mapsto \mathbb{C}, a \in \mathbb{C}$

Definición:

$$\lim_{z \to a} f(z) = L$$

dado $\epsilon > 0$ existe $\delta > 0$ tal que

$$0 < |z - a| < \delta \rightarrowtail |f(z) - L| < \epsilon$$



Límites

$\mathbb{C}$: números complejos

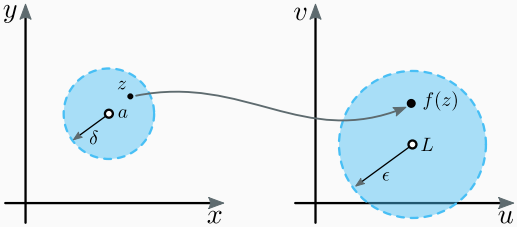
$f : \mathbb{C} \mapsto \mathbb{C}, a \in \mathbb{C}$

Definición:

$$\lim_{z \to a} f(z) = L$$

dado $\epsilon > 0$ existe $\delta > 0$ tal que

$$0 < |z - a| < \delta \rightarrowtail |f(z) - L| < \epsilon$$



Los teoremas usuales sobre límites son válidos. En particular:

Si

$$f(z) = u(x, y) + iv(x, y)$$
$$L = L_1 + iL_2$$
$$a = a_1 + ia_2$$

Entonces:

$$\lim_{z \to a} f(z) = L \longleftrightarrow \begin{cases} \lim_{(x,y) \to (a_1, a_2)} u(x, y) = L_1 \\ \lim_{(x,y) \to (a_1, a_2)} v(x, y) = L_2 \end{cases}$$

Definición:

$$f'(z_0) = \lim_{\Delta z \to 0} \left[\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \right]$$

Derivada

Definición:

$$f'(z_0) = \lim_{\Delta z \to 0} \left[\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \right]$$

Si $w = f(z) = u(x, y) + iv(x, y)$:

$$\begin{aligned} f'(z_0) = \frac{dw}{dz} &= \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} \\ &= \lim_{\Delta z \to 0} \left[\frac{\Delta u + i\Delta v}{\Delta x + i\Delta y} \right] \end{aligned}$$

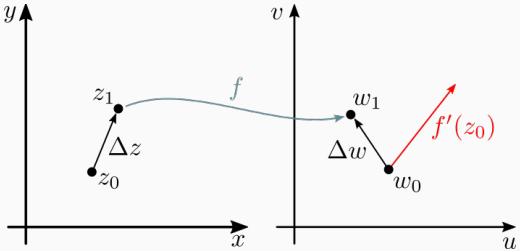
Derivada

Definición:

$$f'(z_0) = \lim_{\Delta z \to 0} \left[\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \right]$$

Si $w = f(z) = u(x, y) + iv(x, y)$:

$$\begin{aligned} f'(z_0) &= \frac{dw}{dz} = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} \\ &= \lim_{\Delta z \to 0} \left[\frac{\Delta u + i\Delta v}{\Delta x + i\Delta y} \right] \end{aligned}$$



Derivada: casos especiales

Caso 1: $\Delta y \equiv 0$.

$$\begin{aligned}\therefore f'(z_0) &= \lim_{\Delta x \to 0} \left[\frac{\Delta u}{\Delta x} + i \frac{\Delta v}{\Delta x} \right] \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \bigg|_{z_0 = (x_0, y_0)}\end{aligned}$$

Derivada: casos especiales

Caso 1: $\Delta y \equiv 0$.

$$\begin{aligned}\therefore f'(z_0) &= \lim_{\Delta x \to 0} \left[\frac{\Delta u}{\Delta x} + i \frac{\Delta v}{\Delta x} \right] \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \bigg|_{z_0 = (x_0, y_0)}\end{aligned}$$

Caso 2: $\Delta x \equiv 0$.

$$\begin{aligned}\therefore f'(z_0) &= \lim_{\Delta y \to 0} \left[\frac{\Delta u}{i \Delta y} + \frac{\Delta v}{\Delta y} \right] = \frac{\partial v}{\partial y} + \frac{1}{i} \frac{\partial u}{\partial y} \bigg|_{z_0 = (x_0, y_0)} \\ &= \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \bigg|_{z_0 = (x_0, y_0)}\end{aligned}$$

Derivada: casos especiales

Caso 1: $\Delta y \equiv 0$.

$$\therefore f'(z_0) = \lim_{\Delta x \to 0} \left[\frac{\Delta u}{\Delta x} + i \frac{\Delta v}{\Delta x} \right]$$
$$= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \bigg|_{z_0 = (x_0, y_0)}$$

Caso 2: $\Delta x \equiv 0$.

$$\therefore f'(z_0) = \lim_{\Delta y \to 0} \left[\frac{\Delta u}{i \Delta y} + \frac{\Delta v}{\Delta y} \right] = \frac{\partial v}{\partial y} + \frac{1}{i} \frac{\partial u}{\partial y} \bigg|_{z_0 = (x_0, y_0)}$$
$$= \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \bigg|_{z_0 = (x_0, y_0)}$$

Ecuaciones de Cauchy-Riemann

Si $f = u + iv$ es diferenciable (analítica), entonces:

$$u_x = v_y$$
$$u_y = -v_x$$

Derivada: ejemplo 1

Ecuaciones de Cauchy-Riemann:

$$\begin{aligned} f(z) &= z^2 = (x+iy)^2 \\ &= (x^2 - y^2) + i(2xy) \end{aligned}$$

$$\left.\begin{aligned} u_x &= 2x, \quad & v_x &= 2y \\ u_y &= -2y, \quad & v_y &= 2x \end{aligned}\right\} \Rightarrow \begin{aligned} u_x &= v_y \\ u_y &= -v_x \end{aligned}$$

$$f(z) \mapsto \text{diferenciable}$$

Derivada: ejemplo 1

Ecuaciones de Cauchy-Riemann:

$$
\begin{aligned}
f(z) &= z^2 = (x+iy)^2 \\
&= (x^2 - y^2) + i(2xy)
\end{aligned}
$$

$$
\left.\begin{aligned}
u_x &= 2x, \quad & v_x &= 2y \\
u_y &= -2y, \quad & v_y &= 2x
\end{aligned}\right\} \Rightarrow \begin{aligned} u_x &= v_y \\ u_y &= -v_x \end{aligned}
$$

$f(z) \mapsto$ diferenciable

Derivada por definición:

$$
\begin{aligned}
\frac{f(z_0+\Delta z) - f(z_0)}{\Delta z} &= \frac{(z_0+\Delta z)^2 - z_0^2}{\Delta z} \\
&= \frac{2z_0\Delta z + \Delta z^2}{\Delta z} \qquad (\Delta z \neq 0) \\
&= 2z_0 + \Delta z
\end{aligned}
$$

$$
\therefore \quad \boxed{f'(z_0) = 2z_0}
$$

Derivada: ejemplo 2

$$f(z) = \bar{z} = x - iy$$
$$u = x,\ v = -y \Rightarrow u_x \neq v_y$$

Derivada: ejemplo 2

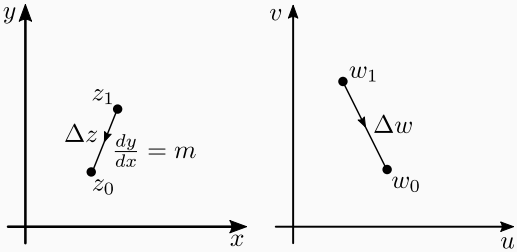
$$f(z) = \bar{z} = x - iy$$
$$u = x,\ v = -y \Rightarrow u_x \neq v_y$$

$$\frac{\Delta w}{\Delta z} = \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$$
$$= \frac{1 - i\dfrac{\Delta y}{\Delta x}}{1 + i\dfrac{\Delta y}{\Delta x}} \to \frac{1 - \dfrac{dy}{dx}}{1 + i\dfrac{dy}{dx}}$$

Derivada: ejemplo 2

$$f(z) = \bar{z} = x - iy$$
$$u = x,\ v = -y \Rightarrow u_x \neq v_y$$

$$\frac{\Delta w}{\Delta z} = \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$$
$$= \frac{1 - i\dfrac{\Delta y}{\Delta x}}{1 + i\dfrac{\Delta y}{\Delta x}} \to \frac{1 - \dfrac{dy}{dx}}{1 + i\dfrac{dy}{dx}}$$



$$\frac{\Delta w}{\Delta z} = \frac{1 - im}{1 + im} = g(m)$$

Aplicación: ecuación de Laplace y ejemplo

$u(x, y)$ satisface la ecuación de Laplace si:

$$\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$$

Aplicación: ecuación de Laplace y ejemplo

$u(x, y)$ satisface la ecuación de Laplace si:

$$\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$$

Si $u + iv$ es **analítica**:

$$u_x = v_y \ \therefore \ u_{xx} = v_{yx}$$
$$u_y = -v_x \ \therefore \ u_{yy} = -v_{xy}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$
$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Aplicación: ecuación de Laplace y ejemplo

$u(x, y)$ satisface la ecuación de Laplace si:

$$\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$$

Si $u + iv$ es **analítica**:

$$u_x = v_y \;\therefore\; u_{xx} = v_{yx}$$
$$u_y = -v_x \;\therefore\; u_{yy} = -v_{xy}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$
$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Ejemplo:

$$f(z) = z^2 \longrightarrow \begin{cases} u = x^2 - y^2 \\ v = 2xy \end{cases}$$

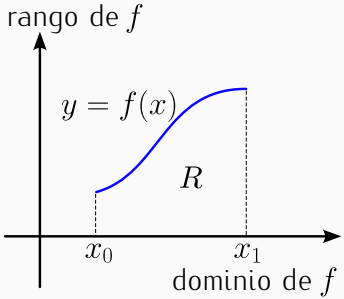
$$\therefore \left.\begin{aligned} u_{xx} + u_{yy} &= 0 \\ v_{xx} + v_{yy} &= 0 \end{aligned}\right\}$$

Pausa para resolver problemas: 9 – 13.

Revisión:

$$\int_{x_0}^{x_1} f(x)\,dx = \lim_{\substack{\text{máx} \\ \Delta x \to 0}} \sum_{k=1}^{n} f(c_k^*)\Delta x_k$$

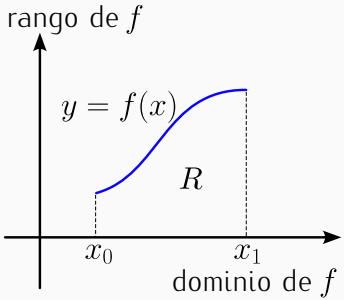
$$= F(x_1) - F(x_0),\ F' = f$$



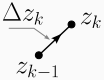
Revisión:

$$\int_{x_0}^{x_1} f(x)\,dx = \lim_{\substack{\text{máx}\\ \Delta x\to 0}} \sum_{k=1}^{n} f(c_k^*)\Delta x_k$$

$$= F(x_1) - F(x_0),\ F' = f$$

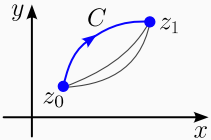


$$\int_{z_0}^{z_1} f(z)\,dz \overset{?}{=} \lim_{\substack{\text{máx}\\ \Delta z\to 0}} \sum_{k=1}^{n} f(c_k^*)\overset{?}{\Delta z_k}$$



$$C: \begin{cases} x = x(t) \\ y = y(y) \end{cases} = \begin{cases} \vec{R} = x(t)\hat{\imath} + y(t)\hat{\jmath} \\ z = x(t) + iy(t) \end{cases}$$

$$t_0 \le t \le t_1$$



$$\int_{C:z_0}^{z_1} f(z)\,dz = \lim_{\substack{\text{máx}\\ \Delta z\to 0}} \sum_{k=1}^{n} f(c_k(t_k))\frac{\Delta z_k}{\Delta t_k}\Delta t_k$$

$$\therefore \int_{C:z_0}^{z_1} f(z)\,dz = \int_{t_0}^{t_1} f(z(t))z'(t)\,dt$$

En términos de u y v: $f(z) = u + iv$,
$\Delta z = \Delta x + i\Delta y$:

$$\int_{\substack{C \\ z_0}}^{z_1} f(z)\,dz = \int_{\substack{C \\ (x_0,y_0)}}^{(x_1,y_1)} (u+iv)(dx+idy) =$$

$$\int_{\substack{C \\ (x_0,y_0)}}^{(x_1,y_1)} (u\,dx - v\,dy) + i\int_{\substack{C \\ (x_0,y_0)}}^{(x_1,y_1)} (v\,dx + u\,dy)$$

$$C : \begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

Si $u + iv$ es analítica: $u_x = v_y,\ u_y = -v_x$.

$$\therefore \left.\begin{matrix} u\,dx - v\,dy \\ v\,dx + u\,dy \end{matrix}\right\} \text{ es diferencial exacta.}$$

$\therefore$ Si $f = u + iv$ en analítica:

$$\int_{z_0}^{z_1} f(z)\,dz$$

es **independiente** de C, y

$$\boxed{\oint_C f(z)\,dz = 0, \quad \forall C}$$

f analítica $\rightarrow$

$$\int_{z_0}^{z_1} f(z)\,dz = F(z_1) - F(z_0), \quad F' = f$$

Nota:

$$\oint_C f(z)\,dz$$

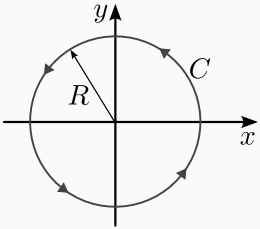
no necesariamente es 0 si f no es analítica.

Ejemplo

Calcular:

$$\oint_C \frac{dz}{z}$$

donde



El integrando es analítico en $\mathbb{C}$ excepto en $z = 0$.

Método #1:

$$\oint_C \frac{dz}{z} = \oint_C \frac{dx + idy}{x + iy}$$

$$= \oint_C \frac{(x - iy)(dx + idy)}{x^2 + y^2} =$$

$$\oint_C \frac{x\,dx + y\,dy}{x^2 + y^2} + i \oint_C \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

En C: $x = R\cos\theta$, $y = R\operatorname{sen}\theta$,
$dx = -R\operatorname{sen}\theta\,d\theta$,
$dy = R\cos\theta\,d\theta$,
$x^2 + y^2 = R^2$, $0 \leq \theta \leq 2\pi$.

$$\therefore \oint_C \frac{dz}{z} = 0$$

$$+ i \int_0^{2\pi} \frac{R^2(\operatorname{sen}^2\theta + \cos^2\theta)\,d\theta}{R^2}$$

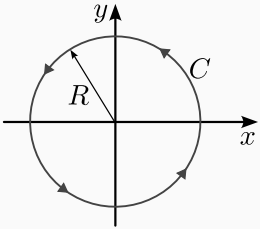
$$= \boxed{2\pi i}$$

Ejemplo

Calcular:

$$\oint_C \frac{dz}{z}$$

donde



El integrando es analítico en $\mathbb{C}$ excepto en $z = 0$.

Método #1:

$$\oint_C \frac{dz}{z} = \oint_C \frac{dx + idy}{x + iy}$$

$$= \oint_C \frac{(x - iy)(dx + idy)}{x^2 + y^2} =$$

$$\oint_C \frac{x\,dx + y\,dy}{x^2 + y^2} + i \oint_C \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

En C: $x = R\cos\theta$, $y = R\operatorname{sen}\theta$,
$dx = -R\operatorname{sen}\theta\,d\theta$,
$dy = R\cos\theta\,d\theta$,
$x^2 + y^2 = R^2$, $0 \leq \theta \leq 2\pi$.

$$\therefore \oint_C \frac{dz}{z} = 0$$

$$+ i \int_0^{2\pi} \frac{R^2(\operatorname{sen}^2\theta + \cos^2\theta)\,d\theta}{R^2}$$

$$= \boxed{2\pi i}$$

Método #2:

$C : z = Re^{i\theta},\ 0 \leq \theta \leq 2\pi.$

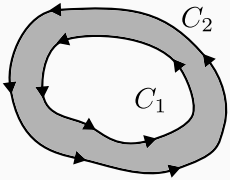
$$\frac{dz}{d\theta} = iRe^{i\theta}$$

$$\oint_C \frac{dz}{z} = \int_0^{2\pi} \frac{1}{z(\theta)} \frac{dz}{d\theta} d\theta$$

$$= \int_0^{2\pi} \frac{iRe^{i\theta}}{Re^{i\theta}} d\theta$$

$$= \boxed{2\pi i}$$

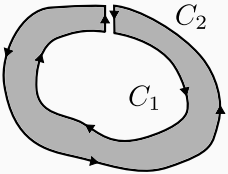
Geometría elástica (topología):



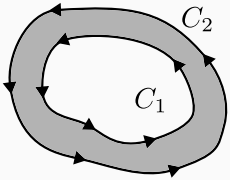
$$\oint_{\phi} f(z)\,dz = \oint_{C_2} f(z)\,dz - \oint_{C_1} f(z)\,dz = 0$$

Si f es analítica en C_1 y C_2, y en la región entre ellas, entonces:

$$\oint_{C_1} f(z)\,dz = \oint_{C_2} f(z)\,dz$$

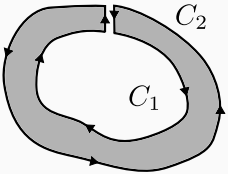


Geometría elástica (topología):



Si f es analítica en C_1 y C_2, y en la región entre ellas, entonces:

$$\oint_{C_1} f(z)\,dz = \oint_{C_2} f(z)\,dz$$

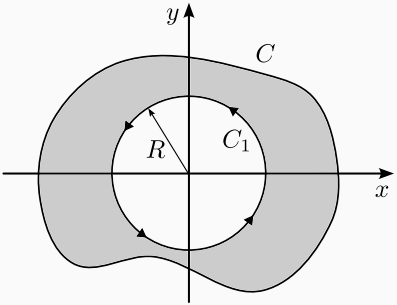


$$\oint_{\not C} f(z)\,dz = \oint_{C_2} f(z)\,dz - \oint_{C_1} f(z)\,dz = 0$$

Ejemplo: calcular

$$\oint_C \frac{dz}{z}$$

donde



$$\oint_C \frac{dz}{z} = \oint_{C_1} \frac{dz}{z} = 2\pi i$$

Pausa para resolver problemas: 14 – 17.

$$e^z = ?, \qquad \operatorname{sen} z = ?$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = ?$$

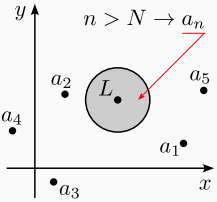
Definición : .

$\lim\limits_{n\to\infty} a_n = L$ significa que

dado $\varepsilon > 0$, ($\varepsilon \in \mathbb{R}$), existe N, tal que

$$n > N \to |a_n - L| < \varepsilon$$

Gráficamente:



En forma similar podemos definir:

$$\sum_{n=1}^{\infty} c_n = \lim_{n\to\infty} (c_1 + \cdots + c_n)$$

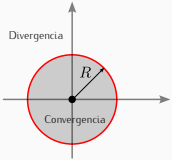
Por su estructura, **son válidos** todos los teoremas usuales.

En particular, si $S = \{z : \sum a_n z^n \text{ converge }\}$, entonces pueden darse los siguientes casos:

i) $S = \{0\}$.

ii) $S = \mathbb{C}$ (todos los números complejos).

iii) Existe un $R > 0$ tal que $S = \{z : |z| < R\}$, y la convergencia es absoluta e uniforme para $|z| \leq r < R$.



Entonces podemos **definir**:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$$\operatorname{sen} z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$\operatorname{sen} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

Se puede entonces probar que:

$$e^{iz} = \cos z + i \operatorname{sen} z$$

o

$$e^{ix} = \cos x + i \operatorname{sen} x$$

$$\therefore (r, \theta) = r\cos\theta + ir \operatorname{sen}\theta$$
$$= re^{i\theta}$$

Tres observaciones:

1. $z = re^{i\theta} = re^{i(\theta+2\pi k)}$

$$\therefore \log z = \log r + \log e^{i(\theta+2\pi k)}$$
$$= \ln r + i(\theta + 2\pi k)$$

$\therefore \log z$ es multivaluada, el **valor principal** es $-\pi < \theta \leq \pi$.

2.

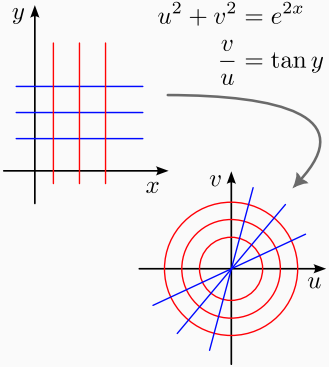
$$\cosh ix = \frac{e^{ix} + e^{-ix}}{2} = \frac{(\cos x + i \operatorname{sen} x)}{2} + \frac{\cos(-x) + i \operatorname{sen}(-x)}{2} = \cos x$$

3. $e^z = e^{x+iy} = e^x e^{iy} =$

$$e^x(\cos y + i \operatorname{sen} y) =$$
$$e^x \cos y + ie^x \operatorname{sen} y$$
$$u(x, y) + iv(x, y)$$

u y v representan un **mapeo conforme** real:

$$u^2 + v^2 = e^{2x}$$
$$\frac{v}{u} = \tan y$$



Entonces podemos **definir**:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$$\operatorname{sen} z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$\operatorname{sen} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

Se puede entonces probar que:

$$e^{iz} = \cos z + i \operatorname{sen} z$$

o

$$e^{ix} = \cos x + i \operatorname{sen} x$$

$$\therefore (r, \theta) = r\cos\theta + ir \operatorname{sen}\theta$$
$$= re^{i\theta}$$

Tres observaciones:

1. $z = re^{i\theta} = re^{i(\theta+2\pi k)}$

$$\therefore \log z = \log r + \log e^{i(\theta+2\pi k)}$$
$$= \ln r + i(\theta + 2\pi k)$$

$\therefore \log z$ es multivaluada, el **valor principal** es $-\pi < \theta \leq \pi$.

2.

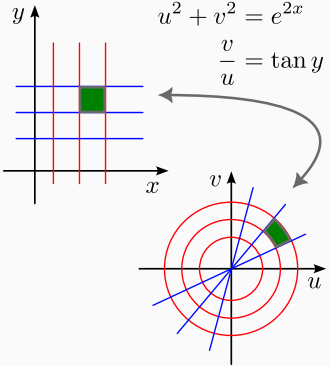
$$\cosh ix = \frac{e^{ix} + e^{-ix}}{2} = \frac{(\cos x + i \operatorname{sen} x)}{2} + \frac{\cos(-x) + i \operatorname{sen}(-x)}{2} = \cos x$$

3. $e^z = e^{x+iy} = e^x e^{iy} =$

$$e^x(\cos y + i \operatorname{sen} y) =$$
$$e^x \cos y + ie^x \operatorname{sen} y$$
$$u(x, y) + iv(x, y)$$

u y v representan un **mapeo conforme** real:

$$u^2 + v^2 = e^{2x}$$
$$\frac{v}{u} = \tan y$$



Aplicación a series "reales"

$$\frac{1}{1-u} = 1 + u + u^2 + \cdots$$
$$= \sum_{n=0}^{\infty} u^2$$

converge para $|u| < 1$.

$$\therefore \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \cdots$$
$$= \sum_{n}^{\infty} (-1)^n x^{2n}$$

converge para $|x| < 1$.

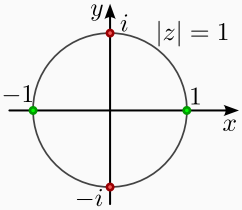
¿Qué pasa en $x = \pm 1$?

$$\sum_{n=0}^{\infty} (-1)^n z^{2n} = \frac{1}{1+z^2}, \ (|z| < 1)$$

$$\frac{1}{1+z^2} = \frac{1}{(z+i)(z-i)}$$

$\therefore z = \pm i \leftarrow$ ¡problema!

Gráficamente:



Los puntos problemáticos están **sobre** $|z| = 1$, pero no en $z = 1$ o en $z = -1$.

Pausa para resolver problemas: 18 – 19.

Lecturas recomendadas

- E. Kreyszig, H. Kreyszig y E.J. Norminton. ***Advanced Engineering Mathematics.*** Hoboken, USA: John Wiley & Sons, Inc, 2011. Capítulo 13 – 16.
- M.R. Spiegel et al. ***Variable compleja.*** Mexico: McGraw-Hill, 1991. Capítulos 1, 2, 6, 14.